Hawraa Moamin

M.Sc. Computer Engineering (Software) · Computer Vision & Trustworthy AI

Computer-vision researcher (state-space models, biometrics, remote sensing) moving toward the **security and trustworthiness of agentic AI**: what agents see, who they act for, and how that can be verified.

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SIGNED CV
Scan to verify

2
peer-reviewed journal papers, Franklin Open (2025, 2026)

4.00/4.00
M.Sc. CGPA; B.Sc. CGPA 3.82

RESEARCH DIRECTION

Multimodal attack surface of agents
How images, screenshots and on screen text can steer computer use agents (visual prompt injection, adversarial content), and how to detect or contain it.

Private identity for delegation
Letting an agent prove it acts for a specific, consenting human (biometrics, liveness) without exposing the biometric data, via FHE or secure computation.

Efficient vision for private inference
Small, strong models (e.g., Vision Mamba at ~7M params) that make on device or encrypted inference practical for agents that see.

PUBLICATIONS

[1] A. S. Altaher, **H. Moamin**, M. Khalid, A. Altaher, C. Bang, H. Altaher, M. T. Jan, H. Zhuang. **"Vision Mamba's superior performance in aerial scene classification: An evaluation across datasets."** *Franklin Open*, vol. 16, 100643, 2026.
PUBLISHED doi.org/10.1016/j.fraope.2026.100643 · Role: conceptualization, methodology, writing

[2] C. Bang, A. S. Altaher, A. Altaher, **H. Moamin**, T. Alshammari, B. Alkanjr, H. Altaher, M. G. Al-Jassani, H. Zhuang. **"Finger knuckle print classification: Leveraging vision Mamba for low complexity and high accuracy."** *Franklin Open*, 100261, 2025.
PUBLISHED doi.org/10.1016/j.fraope.2025.100261 · Role: conceptualization, methodology, writing

[3] **Real-time American Sign Language translation combining signal and image processing, with a mobile app interface.**
IN PREPARATION Deep learning pipeline fusing signal and image based features for live translation, delivered through an app.

RESEARCH PROJECTS

Aerial scene classification with Vision Mamba Feb 2024 – May 2024
Prof. Hanqi Zhuang's group, EECS, Florida Atlantic University
- Benchmarked Vision Mamba against established CNN and transformer models on AID (30 classes) and NWPU-RESISC45 (45 classes): **96.59%** and **94.81%** with ~7M parameters (ResNet50: 25M; ViT: 86M); handled AID's class imbalance. Published as [1].

Finger-knuckle-print biometric recognition Jan 2023 – Mar 2023
Prof. Hanqi Zhuang's group, EECS, Florida Atlantic University
- Evaluated Vision Mamba against 17 pretrained networks on the PolyU FKP dataset (7,920 images, 165 subjects): **99.1%** accuracy vs. AlexNet 96.2%, EfficientNet 98.0%, SCNN 98.3%; robust when parts of the knuckle are occluded. Published as [2].

EXPERIENCE

Engineer Oct 2024 – Sep 2026
Ministry of Education, Iraq
- Programming projects; process optimization and new technical solutions.

Lecturer April 2024 – Oct 2024
Sadr Iraq University College, Iraq
- Lectures and workshops on programming languages.

Lecturer Feb 2024 – April 2024
Ashur University, Iraq
- Lectures and workshops on digital electronics and engineering concepts.

EDUCATION

M.Sc. Computer Engineering (Software) 2020 – 2021
Imam Reza University, Iran · CGPA 4.00 / 4.00
Thesis: COVID-19 patient diagnosis with CPDS features and KNN classification

B.Sc. Computer Engineering 2015 – 2019
Al-Mustansiriya University, Iraq · CGPA 3.82

SKILLS

Deep learning & vision: Image classification, Vision Mamba / state space models, vision transformers, transfer learning, biometrics, remote sensing, class imbalance

Security & cryptography (foundations): hash functions, digital signatures, signed attestations, public key distribution. See the signed CV at hawraa-moamin.github.io

Programming: Python, C++, MATLAB, JavaScript, Visual Basic; Linux and Windows

Data: data analysis, data engineering, tabular prediction

Teaching: university lecturing; workshops for data science and developer communities

Languages: Arabic, English, Persian